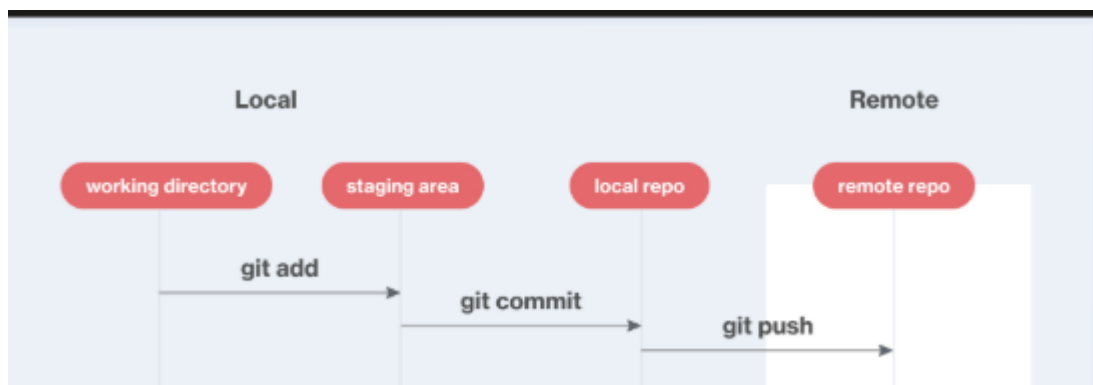


Git and GitHub Introduction

Class no:1

Git

- **Definition:** Git is a version control system created by Linus Torvalds (the creator of Linux).
- **Purpose:** It helps developers track changes in their code, collaborate with others, and roll back to previous versions if needed.
- **Key Features:**
 - Keeps a history of all changes.
 - Allows branching (working on new features separately).
 - Enables merging (combining changes back together).
 - Works locally on your computer — you don't need the internet to use Git.



Architecture diagram of git

GitHub

- **Definition:** GitHub is a cloud-based hosting service for Git repositories.
- **Purpose:** It makes collaboration easier by providing a platform where developers can share their Git projects online.
- **Key Features:**
 - Hosts repositories so others can access them.
 - Provides tools for collaboration (issues, pull requests, code reviews).
 - Integrates with CI/CD pipelines for automated testing and deployment.
 - Adds a social layer — developers can follow projects, star repositories, and contribute to open source.

Difference between Git and GitHub

Feature	Git	GitHub
Type	Version control system	Web-based hosting service
Function	Tracks and manages code locally	Hosts Git repositories online
Accessibility	Local machine, offline capable	Requires internet for collaboration
Collaboration	Manual sharing via email, server	Easy collaboration, pull requests, issues
Main Use Case	Code versioning	Collaborative project management

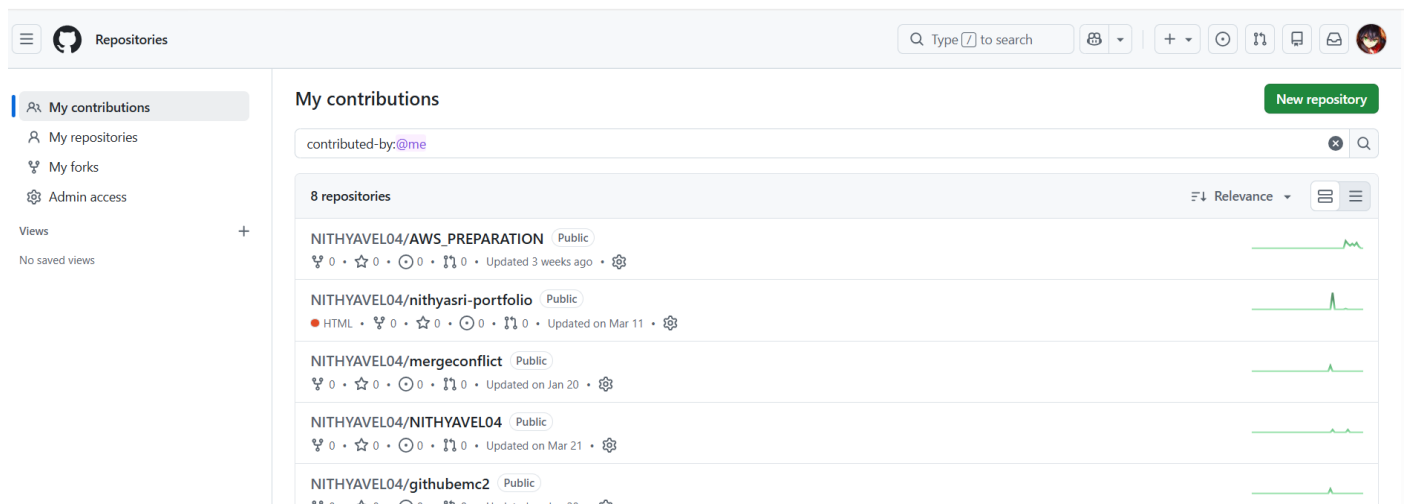
Git is the tool that manages and tracks changes in your code.

GitHub is a platform that lets you host Git repositories online and work collaboratively.

Together, they provide a seamless workflow for modern software development.

Hands on: Creating a repository in gitHub and implementing basic git commands

Step no: 1 Login to your GitHub account → Choose **New repository**.



Name: mynewreporforpracticegit

Type: Public

Enable: Readme

(.git ignore is used to store files that doesn't requires any tracking. So don't enable it.)

Create repository.

Once created → click **code** → copy the **https web url**

main 1 Branch 0 Tags

Go to file

Add file

<> Code

NITHYAVEL04 Initial commit

README.md

Initial commit

README

mynewrepoforpracticegit

Local

Codespaces

Clone

?

HTTPS

SSH

GitHub CLI

<https://github.com/NITHYAVEL04/mynewrepoforprac>

Copy

Clone using the web URL.

Open with GitHub Desktop

Download ZIP

Step no:2 Now open the git bash in our system, to clone the remote repo to our local repo use the following commands.

cd downloads → to get inside of downloads, where we are going to clone our remote repo

git clone https_web_url → helps to clone the remote repo in our local repo

cd repository_name → helps to get inside of our repository

ls → shows the list of files

ls -la → **l**: Shows long listing of file with its all information ; **a**: shows all the hidden files as well.

```
nithya@DESKTOP-NBCS2S9 MINGW64 ~  
$ cd downloads
```

```
nithya@DESKTOP-NBCS2S9 MINGW64 ~/downloads  
$ git clone https://github.com/NITHYAVEL04/mynewrepoforpracticegit.git  
Cloning into 'mynewrepoforpracticegit'...  
remote: Enumerating objects: 3, done.  
remote: Counting objects: 100% (3/3), done.  
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)  
Receiving objects: 100% (3/3), done.
```

```
nithya@DESKTOP-NBCS2S9 MINGW64 ~/downloads  
$ cd mynewrepoforpracticegit
```

```
nithya@DESKTOP-NBCS2S9 MINGW64 ~/downloads/mynewrepoforpracticegit (main)  
$ ls  
README.md
```

```
nithya@DESKTOP-NBCS2S9 MINGW64 ~/downloads/mynewrepoforpracticegit (main)  
$ ls -la  
total 261  
drwxr-xr-x 1 nithya 197121 0 May 11 16:56 ./  
drwxr-xr-x 1 nithya 197121 0 May 11 16:56 ../  
drwxr-xr-x 1 nithya 197121 0 May 11 16:56 .git/  
-rw-r--r-- 1 nithya 197121 25 May 11 16:56 README.md
```

Step no:3 Now we are going to create a file inside it.

touch filename → Creates and empty file

```
nithya@DESKTOP-NBCS2S9 MINGW64 ~/downloads/mynewrepoforpracticegit (main)
$ touch app1.py

nithya@DESKTOP-NBCS2S9 MINGW64 ~/downloads/mynewrepoforpracticegit (main)
$ ls
README.md  app1.py
```

Open the file and add some comments

vim filename → open the file and write the comment

cat filename → To see the contents of the file

```
nithya@DESKTOP-NBCS2S9 MINGW64 ~/downloads/mynewrepoforpracticegit (main)
$ vim app1.py

nithya@DESKTOP-NBCS2S9 MINGW64 ~/downloads/mynewrepoforpracticegit (main)
$ cat app1.py
print("hello world")
```

To know the status of the file → **git status**

We can see it is in untracked files (red colour)

```
nithya@DESKTOP-NBCS2S9 MINGW64 ~/downloads/mynewrepoforpracticegit (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    app1.py

nothing added to commit but untracked files present (use "git add" to track)
```

Also we can use **-s** tag with the status

```
nithya@DESKTOP-NBCS2S9 MINGW64 ~/downloads/mynewrepoforpracticegit (main)
$ git status -s
?? app1.py
```

Step no:4 To bring the file to the staging area use: **git add filename**

Again, check the status, we can see it has been changed to tracked files as shows in green colour.

```
nithya@DESKTOP-NBCS2S9 MINGW64 ~/downloads/mynewrepoforpracticegit (main)
$ git add app1.py

nithya@DESKTOP-NBCS2S9 MINGW64 ~/downloads/mynewrepoforpracticegit (main)
$ git status -s
A  app1.py
```

Step no: 5 Now we are in the staging area, we have to move to the local area. Use: **git commit -m "Our comments"**

-m → stands for messages. It is mandatory in commit.

We can check the status, it will ask to push this into the remote repo.

```
nithya@DESKTOP-NBCS2S9 MINGW64 ~/downloads/mynewrepoforpracticegit (main)
$ git commit -m "My first commit"
[main 4c7ee3f] My first commit
1 file changed, 1 insertion(+)
create mode 100644 appl.py

nithya@DESKTOP-NBCS2S9 MINGW64 ~/downloads/mynewrepoforpracticegit (main)
$ git status
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
(use "git push" to publish your local commits)

nothing to commit, working tree clean
```

git log → Shows all the commits

```
nithya@DESKTOP-NBCS2S9 MINGW64 ~/downloads/mynewrepoforpracticegit (main)
$ git log
commit 4c7ee3fe7306d2f2ba9e6bf9acfccf05db310a1b (HEAD -> main)
Author: NITHYAVEL04 <nithyavel04@gmail.com>
Date: Mon May 11 17:14:27 2026 +0530

    My first commit

commit 2dcb1bb74fdab6ced5810563bc8ba707d463f702 (origin/main, origin/HEAD)
Author: Nithya Sri Palanivel <124020920+NITHYAVEL04@users.noreply.github.com>
Date: Mon May 11 16:49:46 2026 +0530

    Initial commit
```

git log --oneline → Shows all the commits in short hash format

```
nithya@DESKTOP-NBCS2S9 MINGW64 ~/downloads/mynewrepoforpracticegit (main)
$ git log --oneline
4c7ee3f (HEAD -> main) My first commit
2dcb1bb (origin/main, origin/HEAD) Initial commit
```

Step no:6 Now we have push this into the remote repo. Use: **git push origin main**

main: it is the branch inside the repository.

```
nithya@DESKTOP-NBCS2S9 MINGW64 ~/downloads/mynewrepoforpracticegit (main)
$ git push origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 308 bytes | 154.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/NITHYAVEL04/mynewrepoforpracticegit.git
2dcb1bb..4c7ee3f main -> main
```

Step no:7 Open the **git hub account** → go to our **repository** → **refresh it** → we can **see the changes and commits** made in the repository.

Our file has been added there.

The screenshot shows the GitHub interface for a repository named 'mynewrepoforpracticegit' by user 'NITHYAVEL04'. The repository is public and has 1 branch (main) and 0 tags. The commit history shows two commits: 'My first commit' (4c7ee3f) 2 minutes ago and 'Initial commit' (27 minutes ago). The file list shows 'README.md' and 'app1.py'. The README content is 'mynewrepoforpracticegit'.

Create 10 files in local repo and push it in remote repo:

File no:1

```
nithya@DESKTOP-NBCS2S9 MINGW64 ~/downloads/mynewrepoforpracticegit (main)
$ touch myfile1.py

nithya@DESKTOP-NBCS2S9 MINGW64 ~/downloads/mynewrepoforpracticegit (main)
$ git add myfile1.py

nithya@DESKTOP-NBCS2S9 MINGW64 ~/downloads/mynewrepoforpracticegit (main)
$ git commit -m "My first file"
[main cbbe4ff] My first file
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 myfile1.py

nithya@DESKTOP-NBCS2S9 MINGW64 ~/downloads/mynewrepoforpracticegit (main)
$ git push origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 317 bytes | 105.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/NITHYAVEL04/mynewrepoforpracticegit.git
4c7ee3f..cbbe4ff main -> main
```

File no:2

```
nithya@DESKTOP-NBCS2S9 MINGW64 ~/downloads/mynewrepoforpracticegit (main)
$ touch myfile2.py

nithya@DESKTOP-NBCS2S9 MINGW64 ~/downloads/mynewrepoforpracticegit (main)
$ ^C

nithya@DESKTOP-NBCS2S9 MINGW64 ~/downloads/mynewrepoforpracticegit (main)
$ git add myfile2.py

nithya@DESKTOP-NBCS2S9 MINGW64 ~/downloads/mynewrepoforpracticegit (main)
$ git commit -m "My second file"
[main 2ca13a4] My second file
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 myfile2.py

nithya@DESKTOP-NBCS2S9 MINGW64 ~/downloads/mynewrepoforpracticegit (main)
$ git push origin main
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), 272 bytes | 90.00 KiB/s, done.
Total 2 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/NITHYAVEL04/mynewrepoforpracticegit.git
cbbef4ff..2ca13a4 main -> main
```

File no:3

```
nithya@DESKTOP-NBCS2S9 MINGW64 ~/downloads/mynewrepoforpracticegit (main)
$ touch myfile3.py

nithya@DESKTOP-NBCS2S9 MINGW64 ~/downloads/mynewrepoforpracticegit (main)
$ git add myfile3.py

nithya@DESKTOP-NBCS2S9 MINGW64 ~/downloads/mynewrepoforpracticegit (main)
$ git commit -m "My third file"
[main 0025457] My third file
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 myfile3.py

nithya@DESKTOP-NBCS2S9 MINGW64 ~/downloads/mynewrepoforpracticegit (main)
$ git push origin main
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), 240 bytes | 120.00 KiB/s, done.
Total 2 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/NITHYAVEL04/mynewrepoforpracticegit.git
2ca13a4..0025457 main -> main
```

File no:4

```
nithya@DESKTOP-NBCS2S9 MINGW64 ~/downloads/mynewrepoforpracticegit (main)
$ touch myfile4.py

nithya@DESKTOP-NBCS2S9 MINGW64 ~/downloads/mynewrepoforpracticegit (main)
$ git add myfile4.py

nithya@DESKTOP-NBCS2S9 MINGW64 ~/downloads/mynewrepoforpracticegit (main)
$ git commit -m "My fourth file"
[main 023eaaf] My fourth file
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 myfile4.py

nithya@DESKTOP-NBCS2S9 MINGW64 ~/downloads/mynewrepoforpracticegit (main)
$ git push origin main
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), 241 bytes | 120.00 KiB/s, done.
Total 2 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/NITHYAVEL04/mynewrepoforpracticegit.git
0025457..023eaaf main -> main
```

File no:5

```
nithya@DESKTOP-NBCS2S9 MINGW64 ~/downloads/mynewrepoforpracticegit (main)
$ touch myfile5.py

nithya@DESKTOP-NBCS2S9 MINGW64 ~/downloads/mynewrepoforpracticegit (main)
$ git add myfile5.py

nithya@DESKTOP-NBCS2S9 MINGW64 ~/downloads/mynewrepoforpracticegit (main)
$ git commit -m "My fifth file"
[main 73fb40a] My fifth file
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 myfile5.py

nithya@DESKTOP-NBCS2S9 MINGW64 ~/downloads/mynewrepoforpracticegit (main)
$ git push origin main
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), 239 bytes | 239.00 KiB/s, done.
Total 2 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/NITHYAVEL04/mynewrepoforpracticegit.git
023eaaf..73fb40a main -> main
```

File no:6

```
nithya@DESKTOP-NBCS2S9 MINGW64 ~/downloads/mynewrepoforpracticegit (main)
$ touch myfile6.py

nithya@DESKTOP-NBCS2S9 MINGW64 ~/downloads/mynewrepoforpracticegit (main)
$ git add myfile6.py

nithya@DESKTOP-NBCS2S9 MINGW64 ~/downloads/mynewrepoforpracticegit (main)
$ git commit -m "My sixth file"
[main c402da2] My sixth file
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 myfile6.py

nithya@DESKTOP-NBCS2S9 MINGW64 ~/downloads/mynewrepoforpracticegit (main)
$ git push origin main
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), 241 bytes | 120.00 KiB/s, done.
Total 2 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/NITHYAVEL04/mynewrepoforpracticegit.git
73fb40a..c402da2 main -> main
```

File no:7

```
nithya@DESKTOP-NBCS2S9 MINGW64 ~/downloads/mynewrepoforpracticegit (main)
$ touch myfile7.py

nithya@DESKTOP-NBCS2S9 MINGW64 ~/downloads/mynewrepoforpracticegit (main)
$ git add myfile7.py

nithya@DESKTOP-NBCS2S9 MINGW64 ~/downloads/mynewrepoforpracticegit (main)
$ git commit -m "My seventh file"
[main b2e3554] My seventh file
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 myfile7.py

nithya@DESKTOP-NBCS2S9 MINGW64 ~/downloads/mynewrepoforpracticegit (main)
$ git push origin main
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), 244 bytes | 244.00 KiB/s, done.
Total 2 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/NITHYAVEL04/mynewrepoforpracticegit.git
c402da2..b2e3554 main -> main
```

File no:8

```
nithya@DESKTOP-NBCS2S9 MINGW64 ~/downloads/mynewrepoforpracticegit (main)
$ touch myfile8.py

nithya@DESKTOP-NBCS2S9 MINGW64 ~/downloads/mynewrepoforpracticegit (main)
$ git add myfile8.py

nithya@DESKTOP-NBCS2S9 MINGW64 ~/downloads/mynewrepoforpracticegit (main)
$ git commit -m "My eighth file"
[main 31a92a6] My eighth file
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 myfile8.py

nithya@DESKTOP-NBCS2S9 MINGW64 ~/downloads/mynewrepoforpracticegit (main)
$ git push origin main
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), 242 bytes | 242.00 KiB/s, done.
Total 2 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/NITHYAVEL04/mynewrepoforpracticegit.git
b2e3554..31a92a6 main -> main
```

File no:9

```
nithya@DESKTOP-NBCS2S9 MINGW64 ~/downloads/mynewrepoforpracticegit (main)
$ touch myfile9.py

nithya@DESKTOP-NBCS2S9 MINGW64 ~/downloads/mynewrepoforpracticegit (main)
$ git add myfile9.py

nithya@DESKTOP-NBCS2S9 MINGW64 ~/downloads/mynewrepoforpracticegit (main)
$ git commit -m "My ninth file"
[main 8030b08] My ninth file
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 myfile9.py

nithya@DESKTOP-NBCS2S9 MINGW64 ~/downloads/mynewrepoforpracticegit (main)
$ git push origin main
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), 244 bytes | 244.00 KiB/s, done.
Total 2 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/NITHYAVEL04/mynewrepoforpracticegit.git
31a92a6..8030b08 main -> main
```

File no:10

```
nithya@DESKTOP-NBCS2S9 MINGW64 ~/downloads/mynewrepoforpracticegit (main)
$ touch myfile10.py

nithya@DESKTOP-NBCS2S9 MINGW64 ~/downloads/mynewrepoforpracticegit (main)
$ git add myfile10.py

nithya@DESKTOP-NBCS2S9 MINGW64 ~/downloads/mynewrepoforpracticegit (main)
$ git commit -m "My tenth file"
[main d61a4be] My tenth file
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 myfile10.py

nithya@DESKTOP-NBCS2S9 MINGW64 ~/downloads/mynewrepoforpracticegit (main)
$ git push origin main
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), 244 bytes | 244.00 KiB/s, done.
Total 2 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/NITHYAVEL04/mynewrepoforpracticegit.git
8030b08..d61a4be main -> main
```

To delete the local repo : `rm -rf foldername`

```
nithya@DESKTOP-NBCS2S9 MINGW64 ~/downloads/mynewrepoforpracticegit (main)
$ cd ..

nithya@DESKTOP-NBCS2S9 MINGW64 ~/downloads
$ rm -rf mynewrepoforpracticegit/
```

We can see the changes in the remote repo

 mynewrepoforpracticegit Public Pin Watch 0

main 1 Branch 0 Tags T Add file <> Code

 NITHYAVEL04 My tenth file d61a4be · 2 minutes ago 12 Commits

 README.md	Initial commit	1 hour ago
 app1.py	My first commit	1 hour ago
 myfile1.py	My first file	14 minutes ago
 myfile10.py	My tenth file	2 minutes ago
 myfile2.py	My second file	11 minutes ago
 myfile3.py	My third file	10 minutes ago
 myfile4.py	My fourth file	9 minutes ago
 myfile5.py	My fifth file	7 minutes ago
 myfile6.py	My sixth file	6 minutes ago
 myfile7.py	My seventh file	5 minutes ago
 myfile8.py	My eighth file	4 minutes ago
 myfile9.py	My ninth file	3 minutes ago

Steps to delete the remote repo:

1. **Navigate to the Repository:** Open GitHub and go to the main page of the repository you want to delete.
2. **Access Settings:** Under your repository name, click on the "Settings" tab. If you cannot see the "Settings" tab, select the dropdown menu and then click "Settings".
3. **Scroll to Danger Zone:** On the "General" settings page, scroll down to the "Danger Zone" section.
4. **Delete the Repository:** Click on "Delete this repository".
5. **Confirm Deletion**

Danger Zone

Change repository visibility

This repository is currently public.

[Change visibility](#)

Disable branch protection rules

Disable branch protection rules enforcement and APIs

[Disable branch protection rules](#)

Transfer ownership

Transfer this repository to another user or to an organization where you have the ability to create repositories.

[Transfer](#)

Archive this repository

Mark this repository as archived and read-only.

[Archive this repository](#)

Delete this repository

Once you delete a repository, there is no going back. Please be certain.

[Delete this repository](#)

Delete NITHYAVEL04/mynewrepoforpracticegit



NITHYAVEL04/mynewrepoforpracticegit

☆ 0 stars 👁 0 watchers

To confirm, type "NITHYAVEL04/mynewrepoforpracticegit" in the box below

[Delete this repository](#)